Name: Noorain Patankar
Roll no: 5850
SYBCA "A"

PRACTICAL NO 10

1. Create a JSP page to demonstrate the use of BLOB & CLOB.

Using JSP:

Code:

store.jsp

```
<%@ page import="java.sql.*, java.io.*, jakarta.servlet.http.*, jakarta.servlet.annotation.*" %>
<%@ page contentType="text/html;charset=UTF-8" %>
<!DOCTYPE html>
<html>
<body>
<h2>Store Text (CLOB) and Image (BLOB)</h2>
<form method="post" enctype="multipart/form-data">
   Enter Text:<br>
   <textarea name="text" rows="5" cols="40" required></textarea><br><br>
   Select Image:
   <input type="file" name="img" accept="image/*" required><br><br>
   <input type="submit" name="submit" value="Save">
</form>
<%
String submit = request.getParameter("submit");
if(submit != null) {
   Connection con = null;
   PreparedStatement ps = null;
   InputStream imgStream = null;
   try {
      Part filePart = request.getPart("img");
      if(filePart != null && filePart.getSize() > 0) {
         String text = request.getParameter("text");
         imgStream = filePart.getInputStream();
         Class.forName("com.mysql.cj.jdbc.Driver");
         con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/simple_blob_clob?useSSL=false&serverTimezone=UTC", "root", "mamta123");
         ps = con.prepareStatement("INSERT INTO demo(textdata, imgdata) VALUES (?,?)");
         ps.setCharacterStream(1, new StringReader(text));
         ps.setBinaryStream(2, imgStream);
         ps.executeUpdate();
         out.println("<h3 style='color:green;'>✓ Data Stored Successfully!</h3>");
         out.println("<a href='view.jsp'>View Data</a>");
      } else {
         out.println("<h3 style='color:red;'>Please select an image!</h3>");
      }
   } catch(Exception e) {
      out.println("<h3 style='color:red;'>Error: " + e.getMessage() + "</h3>");
   } finally {
      try {
         if(imgStream != null) imgStream.close();
         if(ps != null) ps.close();
         if(con != null) con.close();
      } catch(Exception e) {}
   }}
%>

</body>
</html>
```

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view.jsp

```
<%@ page import="java.sql.*, java.io.*, java.util.Base64" %>
<%@ page contentType="text/html;charset=UTF-8" %>
<html>
<head>
   <style>
      .card { border: 1px solid #ddd; padding: 15px; margin: 10px 0; background: #f9f9f9; }
      img { max-width: 300px; border: 1px solid #ccc; }
   </style>
</head>
<body>
<h2>Display Stored Text & Image</h2>
<a href="store.jsp">Add New Data</a><br><br>
<%
Connection con = null;
Statement st = null;
ResultSet rs = null;
try {
   Class.forName("com.mysql.cj.jdbc.Driver");
   con = DriverManager.getConnection(
      "jdbc:mysql://localhost:3306/simple_blob_clob?useSSL=false&serverTimezone=UTC","root", "mamta123");
   st = con.createStatement();
   rs = st.executeQuery("SELECT * FROM demo ORDER BY id DESC");
   while(rs.next()) {
      out.println("<div class='card'>");
      out.println("<h3>Record ID: " + rs.getInt("id") + "</h3>");
      Reader reader = rs.getCharacterStream("textdata");
      StringBuilder text = new StringBuilder();
      int ch;
      while((ch = reader.read()) != -1) {
         text.append((char)ch);
      }
      out.println("<b>Text:</b><br><p>" + text.toString() + "</p>");
      byte[] imgData = rs.getBytes("imgdata");
      String base64 = Base64.getEncoder().encodeToString(imgData);
      out.println("<b>Image:</b><br>");
      out.println("<img src='" + base64 + "'/>");
      out.println("</div>");
   }
} catch(Exception e) {
   out.println("<h3 style='color:red;'>Error: " + e.getMessage() + "</h3>");
} finally {
   try {
      if(rs != null) rs.close();
      if(st != null) st.close();
      if(con != null) con.close();
   } catch(Exception e) {}
}
%>
</body>
</html>
```

web.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<web-app xmlns="https://jakarta.ee/xml/ns/jakartaee"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="https://jakarta.ee/xml/ns/jakartaee
         https://jakarta.ee/xml/ns/jakartaee/web-app_5_0.xsd"
         version="5.0">
   <servlet>
```

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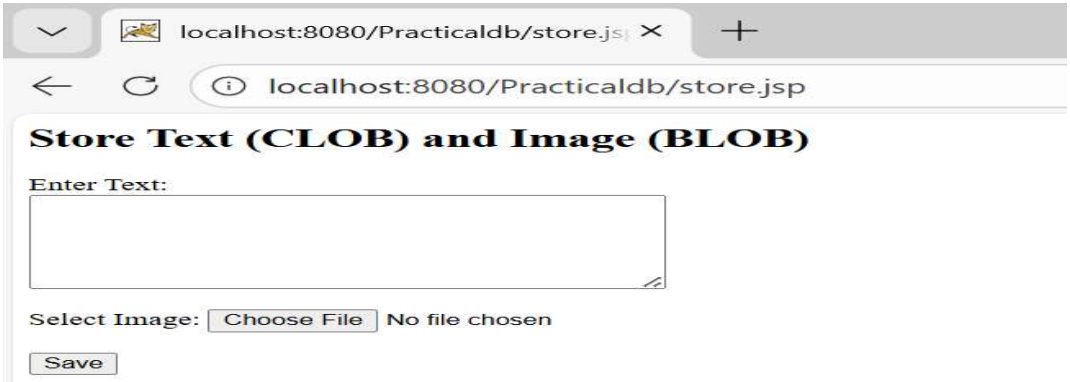
```
        <servlet-name>store</servlet-name>
        <jsp-file>/store.jsp</jsp-file>
        <multipart-config>
            <max-file-size>10485760</max-file-size>
            <max-request-size>20971520</max-request-size>
            <file-size-threshold>0</file-size-threshold>
        </multipart-config>
    </servlet>
    <servlet-mapping>
        <servlet-name>store</servlet-name>
        <url-pattern>/store.jsp</url-pattern>
    </servlet-mapping>
</web-app>
```

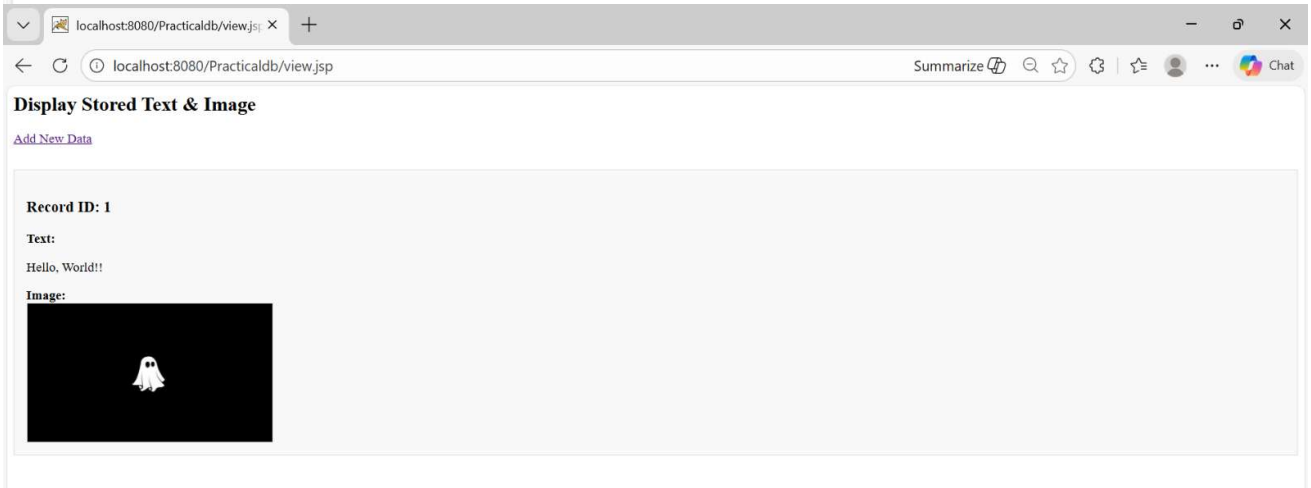
Database and Table:

```
mysql> CREATE DATABASE simple_blob_clob;
Query OK, 1 row affected (0.02 sec)

mysql> USE simple_blob_clob;
Database changed
mysql>
mysql> CREATE TABLE demo (
    ->      id INT AUTO_INCREMENT PRIMARY KEY,
    ->      textdata LONGTEXT,
    ->      imgdata LONGBLOB
    -> );
Query OK, 0 rows affected (0.08 sec)
```

Output:





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Using Servlet:

1. Create a web application using HTML, Servlet, and JDBC to store image files in a MySQL database using the BLOB data type.

Code:

uploadimage.html

```
<!DOCTYPE html>
<html>
<head>
   <title>Upload Image</title>
</head>
<body>
   <h2>Upload Image (blob)</h2>
   <form action="UploadImageServlet" method="post" enctype="multipart/form-data">
      Image name: <input type="text" name="name" required><br><br>
      Select Image: <input type="file" name="img" accept="image/*" required><br><br>
      <input type="submit" value="Upload">
   </form>
</body>
</html>
```

UploadImageServlet.java

```
import java.io.*;

import jakarta.servlet.*;
import jakarta.servlet.annotation.MultipartConfig;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.*;
import java.sql.*;
@WebServlet("/UploadImageServlet")
@MultipartConfig(maxFileSize = 16177215)  // 15MB max file size
public class UploadImageServlet extends HttpServlet {
   protected void doPost(HttpServletRequest request, HttpServletResponse response)
         throws ServletException, IOException {
      response.setContentType("text/html");
      PrintWriter out = response.getWriter();
      Connection con = null;
      PreparedStatement ps = null;
      InputStream inputStream = null;
      try {
         String name = request.getParameter("name");
         Part filePart = request.getPart("img");
         if (name == null || name.trim().isEmpty()) {
            out.println("<h3 style='color:red;'>Error: Please enter an image name!</h3>");
            return;
         }
         if (filePart == null || filePart.getSize() == 0) {
            out.println("<h3 style='color:red;'>Error: Please select an image file!</h3>");
            return;
         }
         inputStream = filePart.getInputStream();
         Class.forName("com.mysql.cj.jdbc.Driver");
         con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/web_blob_clob?useSSL=false&serverTimezone=UTC","root",
            "mamta123");
         ps = con.prepareStatement("INSERT INTO image_store(name, img) VALUES (?, ?)");
         ps.setString(1, name);
         ps.setBlob(2, inputStream);
         int result = ps.executeUpdate();
```

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```
        if (result > 0) {
            out.println("<html><body>");
            out.println("<h2 style='color:green;'>✓ Image Stored Successfully!</h2>");
            out.println("<p><b>Image Name:</b> " + name + "</p>");
            out.println("<p><b>File Size:</b> " + filePart.getSize() + " bytes</p>");
            out.println("<br><a href='upload.html'>Upload Another Image</a>");
            out.println("</body></html>");
        } else {
            out.println("<h3 style='color:red;'>Failed to store image.</h3>");
        }
    } catch (ClassNotFoundException e) {
        out.println("<h3 style='color:red;'>ERROR: MySQL Driver not found!</h3>");
        out.println("<p>Make sure mysql-connector-j jar is in WEB-INF/lib/</p>");
        e.printStackTrace(out);
    } catch (SQLException e) {
        out.println("<h3 style='color:red;'>DATABASE ERROR</h3>");
        out.println("<p><b>Error Code:</b> " + e.getErrorCode() + "</p>");
        out.println("<p><b>SQL State:</b> " + e.getSQLState() + "</p>");
        out.println("<p><b>Message:</b> " + e.getMessage() + "</p>");
        e.printStackTrace(out);
    } catch (Exception e) {
        out.println("<h3 style='color:red;'>ERROR</h3>");
        e.printStackTrace(out);
    } finally {
        try {
            if (inputStream != null) inputStream.close();
            if (ps != null) ps.close();
            if (con != null) con.close();
        } catch (SQLException e) {
            e.printStackTrace(out);
        }
    }
  }
}
```

Database and Table:

```
mysql> CREATE DATABASE web_blob_clob;
Query OK, 1 row affected (0.28 sec)

mysql> USE web_blob_clob;
Database changed
mysql> CREATE TABLE image_store (
    -> id INT AUTO_INCREMENT PRIMARY KEY,
    -> name VARCHAR(50),
    -> img LONGBLOB
    -> );
Query OK, 0 rows affected (1.25 sec)

mysql> CREATE TABLE text_store (
    -> id INT AUTO_INCREMENT PRIMARY KEY,
    -> content LONGTEXT
    -> );
Query OK, 0 rows affected (0.04 sec)
```

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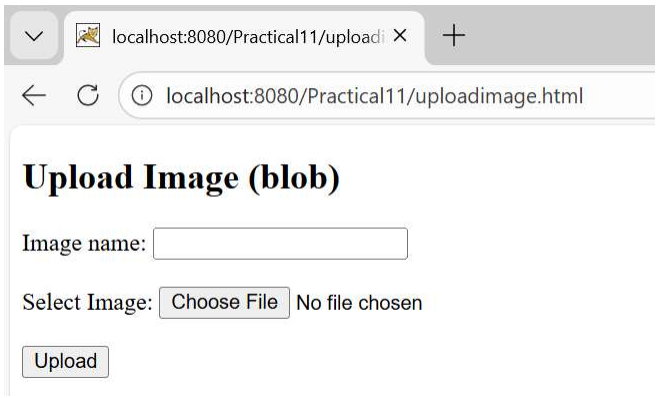
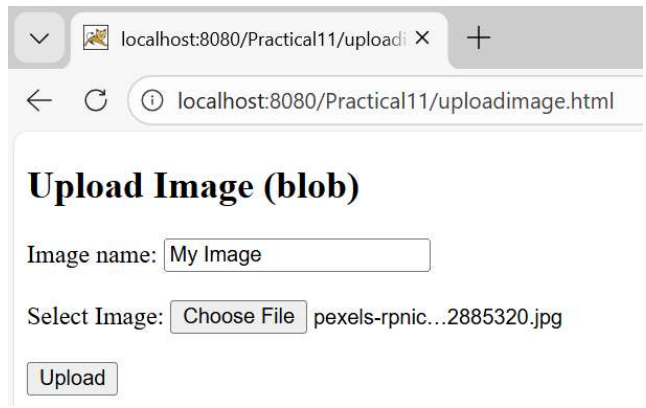
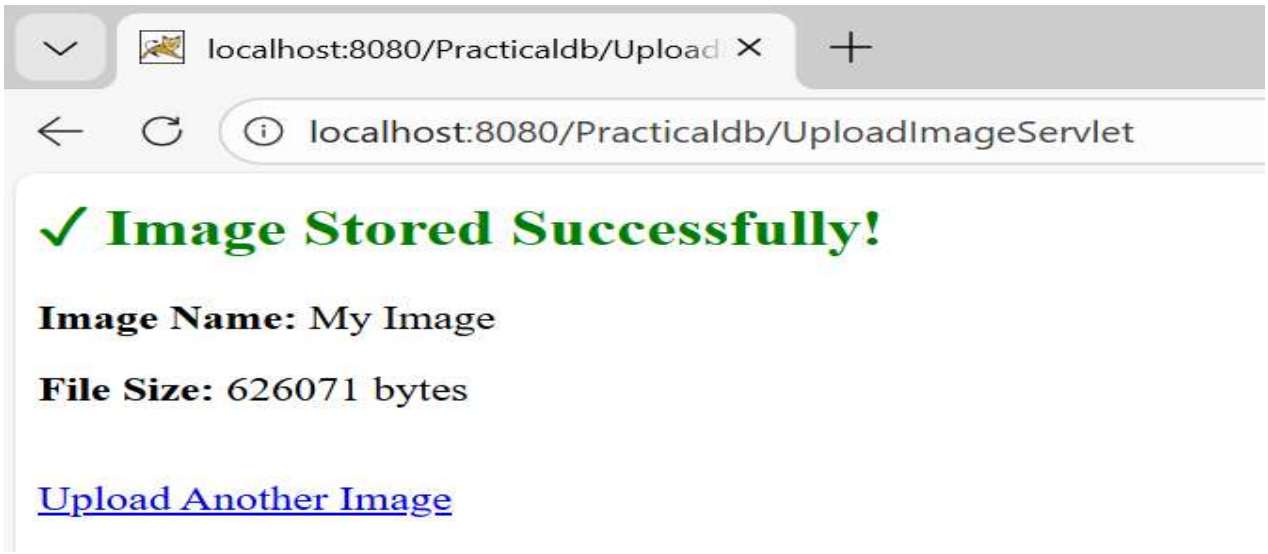
Output:







Database output:

```
mysql> SELECT COUNT(*) as total_images FROM image_store;
+--------------+
| total_images |
+--------------+
|            1 |
+--------------+
1 row in set (0.05 sec)
```

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2. Create a web application using HTML, Servlet, and JDBC to retrieve image files in a MySQL database using the BLOB data type.

Code:

viewImage.html

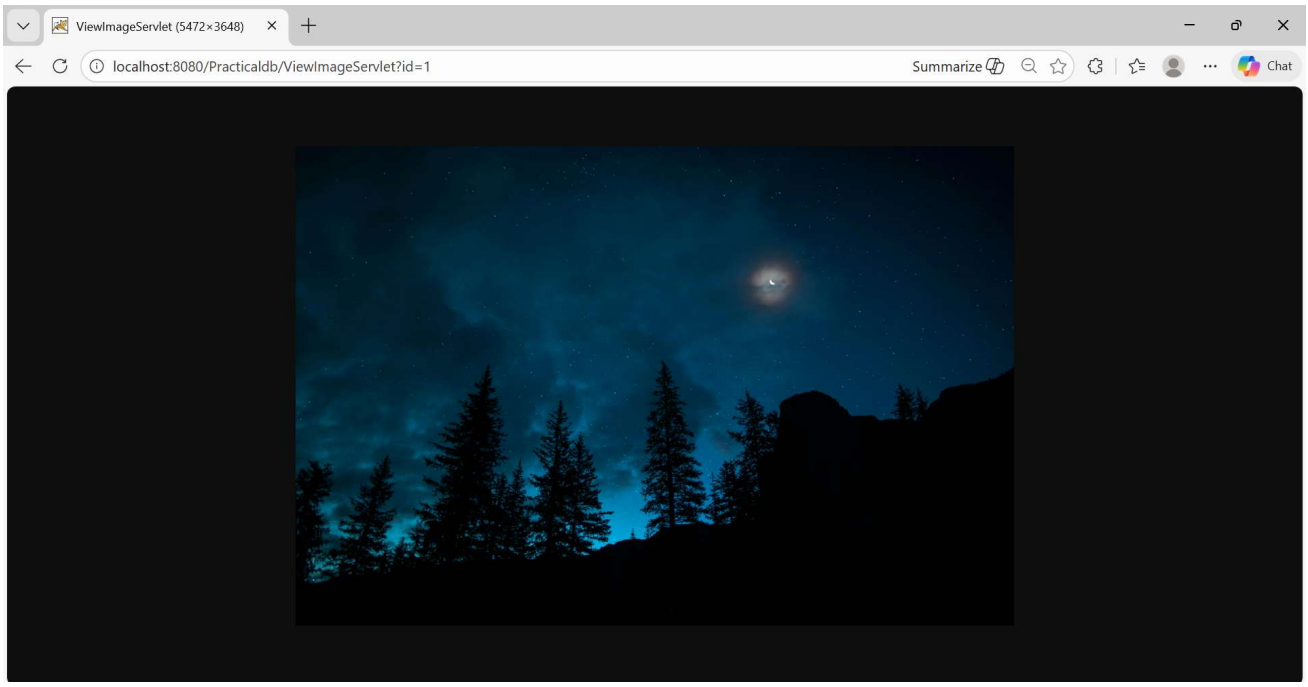
```
<!DOCTYPE html>
<html>
<body>
<h2>View Image</h2>
<form action="ViewImageServlet" method="get">
   Enter Image ID: <input type="text" name="id">
   <input type="submit" value="Show Image">
</form>
</body>
</html>
```

ViewImageServlet.java

```
import java.io.*;
import jakarta.servlet.*;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.*;
import java.sql.*;
@WebServlet("/ViewImageServlet")
public class ViewImageServlet extends HttpServlet {
   protected void doGet(HttpServletRequest request, HttpServletResponse response)
         throws ServletException, IOException {
      String idParam = request.getParameter("id");
      if (idParam == null || idParam.trim().isEmpty()) {
         response.sendError(HttpServletResponse.SC_BAD_REQUEST, "Image ID is required");
         return;
      }
      Connection con = null;
      PreparedStatement ps = null;
      ResultSet rs = null;
      OutputStream os = null;
      try {
         int id = Integer.parseInt(idParam);
         Class.forName("com.mysql.cj.jdbc.Driver");
         con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/web_blob_clob?useSSL=false&serverTimezone=UTC","root",
            "mamta123");
         ps = con.prepareStatement("SELECT img FROM image_store WHERE id=?");
         ps.setInt(1, id);
         rs = ps.executeQuery();
         if (rs.next()) {
            byte[] imgData = rs.getBytes("img");
            response.setContentType("image/jpeg");
            response.setContentLength(imgData.length);
            os = response.getOutputStream();
            os.write(imgData);
            os.flush();
         } else {
            response.sendError(HttpServletResponse.SC_NOT_FOUND,
               "Image with ID " + id + " not found");
         }
      } catch (NumberFormatException e) {
         response.sendError(HttpServletResponse.SC_BAD_REQUEST,
            "Invalid image ID format");
      } catch (ClassNotFoundException e) {
```

```
            response.sendError(HttpServletResponse.SC_INTERNAL_SERVER_ERROR,
                "Database driver not found");
            e.printStackTrace();
        } catch (SQLException e) {
            response.sendError(HttpServletResponse.SC_INTERNAL_SERVER_ERROR,
                "Database error occurred");
            e.printStackTrace();
        } finally {
            try {
                if (rs != null) rs.close();
                if (ps != null) ps.close();
                if (con != null) con.close();
                if (os != null) os.close();
            } catch (SQLException e) {
                e.printStackTrace();
            }
        }
    }
}
```

Output:



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3. Create a web application using HTML, Servlet, and JDBC to store large text data using the CLOB data type.

Code:

uploadText.html

```
<!DOCTYPE html>
<html>
<body>
<h2>Store Large Text (CLOB)</h2>
<form action="UploadTextServlet" method="post">
   Enter Text:<br>
   <textarea name="content" rows="10" cols="50"></textarea><br><br>
   <input type="submit" value="Save">
</form>
</body>
</html>
```

UploadTextServlet.java

```
import java.io.*;

import jakarta.servlet.*;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.*;
import java.sql.*;
@WebServlet("/UploadTextServlet")
public class UploadTextServlet extends HttpServlet {
   protected void doPost(HttpServletRequest request, HttpServletResponse response)
         throws ServletException, IOException {
      response.setContentType("text/html");
      PrintWriter out = response.getWriter();

      String text = request.getParameter("content");

      // Validation
      if (text == null || text.trim().isEmpty()) {
         out.println("<h3 style='color:red;'>Error: Please enter some text!</h3>");
         return;
      }

      Connection con = null;
      PreparedStatement ps = null;

      try {
         Class.forName("com.mysql.cj.jdbc.Driver");
         con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/web_blob_clob?useSSL=false&serverTimezone=UTC",
            "root", "mamta123");

         ps = con.prepareStatement("INSERT INTO text_store(content) VALUES (?)");
         ps.setCharacterStream(1, new StringReader(text));
         ps.executeUpdate();

         out.println("<html><body>");
         out.println("<h2 style='color:green;'>✓ Text Stored Successfully!</h2>");
         out.println("<p>Text length: " + text.length() + " characters</p>");
         out.println("<br><a href='uploadText.html'>Upload More Text</a>");
         out.println(" | <a href='ListTextServlet'>View All Texts</a>");
         out.println("</body></html>");
```

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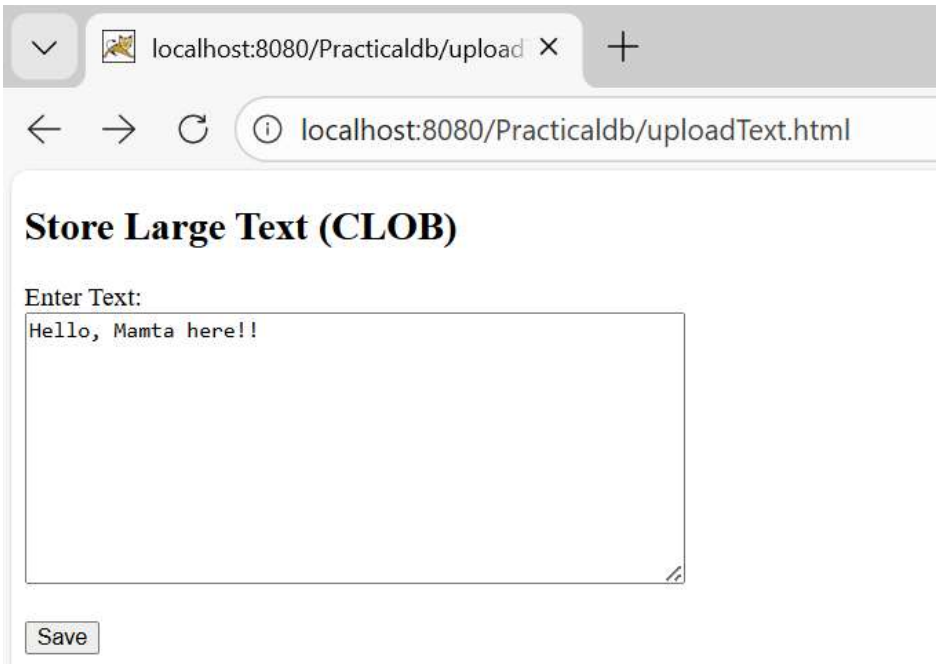
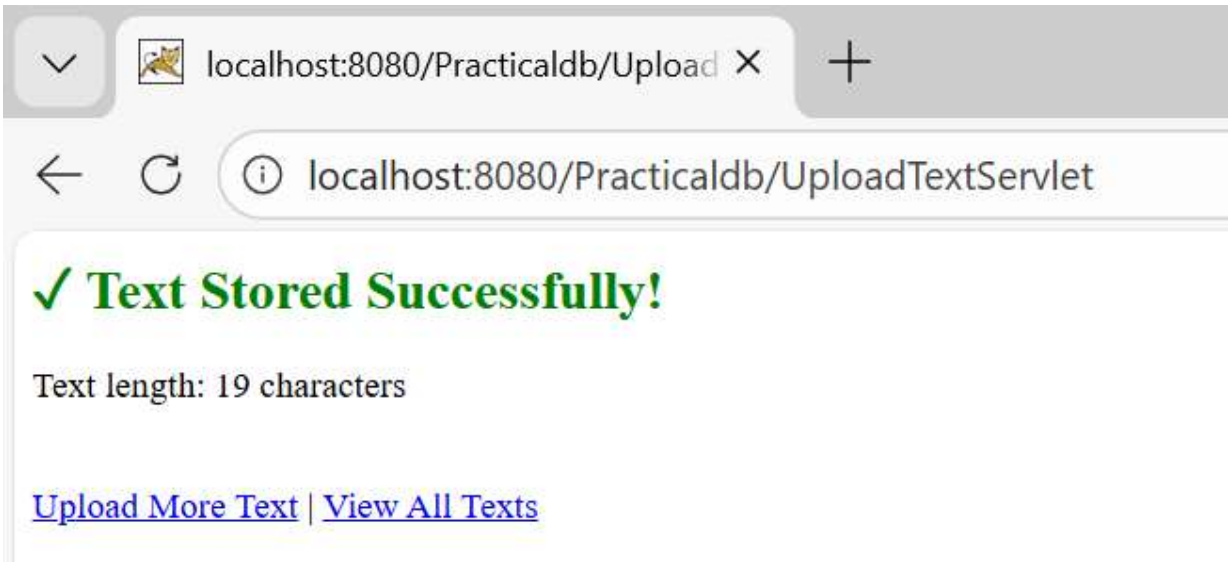
```
        } catch (Exception e) {
            out.println("<h3 style='color:red;'>Error: " + e.getMessage() + "</h3>");
            e.printStackTrace(out);
        } finally {
            try {
                if (ps != null) ps.close();
                if (con != null) con.close();
            } catch (SQLException e) {
                e.printStackTrace();
            }
        }
    }
}
```

Output:





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4. Create a web application using HTML, Servlet, and JDBC to retrieve large text data using the CLOB data type.

Code:

viewText.html

```
<!DOCTYPE html>

<html>
<body>
<h2>View Stored Text</h2>
<form action="viewTextServlet" method="get">
   Enter Text ID: <input type="text" name="id">
   <input type="submit" value="Show Text">
</form>
</body>
</html>
```

viewTextServlet.java

```
import java.io.*;
import jakarta.servlet.*;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.*;
import java.sql.*;

@WebServlet("/viewTextServlet")
public class viewTextServlet extends HttpServlet {
   protected void doGet(HttpServletRequest request, HttpServletResponse response)
         throws ServletException, IOException {

      response.setContentType("text/html");
      PrintWriter out = response.getWriter();

      String idParam = request.getParameter("id");
      if (idParam == null) {
         out.println("Error: ID required");
         return;
      }

      Connection con = null;
      PreparedStatement ps = null;
      ResultSet rs = null;
      Reader reader = null;

      try {
         int id = Integer.parseInt(idParam);

         Class.forName("com.mysql.cj.jdbc.Driver");
         con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/web_blob_clob?useSSL=false&serverTimezone=UTC",
            "root", "mamta123");

         ps = con.prepareStatement("SELECT content FROM text_store WHERE id=?");
         ps.setInt(1, id);
         rs = ps.executeQuery();

         if (rs.next()) {
            out.println("<html><body>");
            out.println("<h2>Text Content (ID: " + id + ")</h2>");
            out.println("<div style='border:1px solid #ddd; padding:15px; background:#f9f9f9;'>");
```

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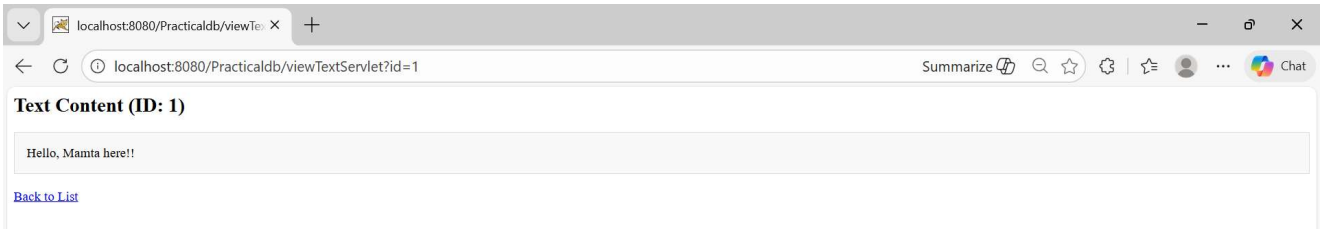
```
            reader = rs.getCharacterStream("content");
            int ch;
            while ((ch = reader.read()) != -1) {
                out.print((char) ch);
            }

            out.println("</div>");
            out.println("<br><a href='ListTextServlet'>Back to List</a>");
            out.println("</body></html>");
        } else {
            out.println("Text not found!");
        }
    } catch (Exception e) {
        out.println("Error: " + e.getMessage());
    } finally {
        try {
            if (reader != null) reader.close();
            if (rs != null) rs.close();
            if (ps != null) ps.close();
            if (con != null) con.close();
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
  }
}
```

Output: